

Module 4

In this module there are 4 assignments.

Assignment 1:

You are required to compute the power of a number by implementing a calculator. Create a class MyCalculator which consists of a single method long power(int, int). This method takes two integers, n and p, as parameters and finds n^p . If either n or p is negative, then the method must throw an exception which says "n or p should not be negative". Also, if both n and p are zero, then the method must throw an exception which says "n and p should not be zero"

Assignment 2:

Create a Java program that performs the following operations on an ArrayList of integers:

1. Initialize an ArrayList with the values `[1, 2, 3, 4, 5]`.
2. Add the value `6` to the end of the ArrayList.
3. Insert the value `10` at index 2.
4. Remove the value at index 3.
5. Print the ArrayList after each operation to demonstrate the changes.

Assignment 3:

Create a Java program that demonstrates the basics of a HashSet:

1. Initialize a HashSet of strings with the values `"apple"`, `"banana"`, `"cherry"`, `"date"`, and `"fig"`.

2. Add the value `""banana""` to the HashSet.
3. Print the size of the HashSet.
4. Use an iterator to print all elements of the HashSet.

Assignment 4:

You are given a list of products. Your task is to complete the following tasks using the Stream API. Write Java code to perform each task and provide the results.

1. Find all products that have a price greater than 1000.
2. Calculate the total price of all products in the list.
3. Find the product with the highest price.
4. Create a list of product names sorted alphabetically.
5. Count the number of products in each category.